

Weekly Research Report — Week 7

Researcher: Harshit Kapur

Summary of Work Done

The backbone model was changed to Claude Opus 5 and the full MedQA evaluation re-run. All knowledge graphs were rebuilt with the stronger extraction model, and PrimeKG — a large-scale authoritative clinical knowledge graph — was integrated. A PubMedQA evaluation was set up and then withdrawn after a data-leakage problem was found.

Backbone model change

The Week 6 negative result left open whether the ceiling was the method or the model. Switching from `llama-3.3-70b-versatile` to `claude-opus-5` tests this directly.

Architecture	Llama 3.3 70B	Claude Opus 5
Baseline, no retrieval	77.1%	96.3%
Flat-vector RAG	76.7%	96.5%
Static GraphRAG	73.4%	95.8%
Hybrid GraphRAG (BM25 + vector + graph)	—	96.2%

The baseline rises 19 points. All architectures now cluster within 0.7 points of each other.

This reframes the Week 6 finding. The earlier result was not that graph retrieval is harmful — it was that a weaker model handles injected context poorly. With a frontier model the benchmark is saturated: 96.3% without any retrieval leaves almost no room for retrieval to contribute. This is a ceiling effect, and it is a more defensible finding than the original negative result.

Hybrid architecture

Following the direction to focus on hybridising vector search with knowledge graphs, a fourth architecture was implemented combining three retrieval sources per query: BM25 keyword search, dense FAISS vector retrieval, and knowledge graph traversal, concatenated as joint context.

Knowledge graphs rebuilt

Graph	Entities	Relations	Source
MedQA	1,330	1,190	500 training documents, Claude Opus 5 extraction
PubMedQA	1,968	1,549	500 abstracts, Claude Opus 5 extraction
PrimeKG	12,752	441,429	Harvard Dataverse, clinically filtered

PrimeKG integrates 20 curated biomedical databases describing 17,080 diseases across 129,375 nodes and 4,050,249 relationships. The full graph was downloaded and filtered to clinical disease and drug relations, giving a graph two orders of magnitude richer than anything extractable from benchmark documents. It provides typed, provenance-tracked relations — disease-phenotype associations, drug contraindications, gene-disease links — that flat document retrieval cannot represent.

PubMedQA evaluation — withdrawn

A PubMedQA evaluation was built and produced an implausible result: the hybrid system scored 100% over the first 130 questions.

Inspection of the retrieved context showed why. The retrieval corpus and the evaluation set were drawn from the same 1,000 labelled instances, so BM25 was retrieving the test question itself, answer included. The system was reading the answer key.

The corpus was re-split, 700 documents for retrieval and 300 held out for evaluation, and the run repeated cleanly. The clean numbers showed the baseline leading, and the evaluation was subsequently dropped from the paper in

favour of the visual QA work begun in Week 8.

Recording this because the failure mode is not obvious — the result looked like success, and only checking the retrieved passages revealed it.

Other evaluations attempted

MedMCQA — set up and abandoned. The test split has `cop = -1` for every row, meaning no answer labels; the dataset is distributed for a competition. The validation split is correctly labelled and was used instead, but the evaluation was dropped when the visual QA direction proved more productive.

Ensemble voting — three retrieval strategies run independently with a majority vote over their answers. Accuracy fell below the baseline. When individual systems are already at 95% and largely agree, voting adds noise rather than removing it. Abandoned.

Plan for Week 8

1. Extend the hybrid architecture to medical visual question answering, where parametric knowledge should be genuinely insufficient
2. Complete PrimeKG-based evaluation on MedQA
3. Run McNemar significance testing across all comparisons
4. Finalise the paper draft